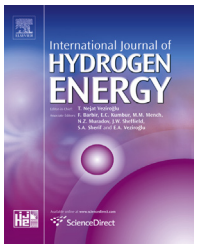
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Comparison of micro-aerobic and anaerobic fermentative hydrogen production from corn straw

Dong Li [a], *Chenxi Jiao* [b], *Wei He* [c], *Zhiying Yan* [a], *Yuexiang Yuan* [a], *Zhidong Li* [a], *Yani Guo* [b], *Xiaofeng Liu* [a,*]

[a] *Key Laboratory of Environmental and Applied Microbiology, Environmental Microbiology Key Laboratory of Sichuan Province, Chengdu Institute of Biology, Chinese Academy of Sciences, Chengdu 610041, China*
[b] *Institute of Environmental and Chemical Engineering, Xi'an Polytechnic University, Xian 710048, China*
[c] *Chengdu Organic Chemicals Co., LTD., Chinese Academy of Sciences, Chengdu 610041, China*

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ABSTRACT

Micro-aerobic and anaerobic fermentative hydrogen productions were carried out in bottle tests (150 mL) and scale-up bioreactors (10 L) using corn straw as a substrate. Hydrogen yield of 41.6 mL/g (volatile solid, VS) obtained from scale-up experiment with optimal oxygen addition of 0.28 mL/(g (total solid, TS)·day) was higher than that of 28.9 mL/g VS without additional oxygen. Oxidation reduction potential was−300~−150 mV for micro-aerobic fermentation, where it was−560~−360 mV for anaerobic fermentation. The micro-aerobic condition had strong buffering ability and could tolerate a greater accumulation of volatile fatty acid. Both micro-aerobic and anaerobic hydrogen production were butyric acid-type fermentation. Microbial community analysis indicated the presence of *Citrobacter* sp. and *Escherichia* sp., which were unique to micro-aerobic system, as well as classical *Enterococcus*, *Bacteroides*, *Peptoclostridium*, *Clostridium* genera that existed in both micro-aerobic and anaerobic fermentation.

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Introduction

Anaerobic fermentative hydrogen production has been successfully used to treat easily degradable carbohydrates such as glucose, xylose, sucrose, lactose, molasses, and starch, reviewed by Davila-Vazquez et al. [1]. Agricultural straw provides abundant raw material for biological hydrogen production, and is mainly composed of cellulose, hemicellulose, and lignin, which are closely intertwined [2]. To date, studies of biological hydrogen production from straw have focused on pretreatment approaches, including steam-explosion [3], ozone [4], white-rot fungal [5], calcium hydroxide [6], hydrochloric acid [7], and sulfuric acid [8] treatments, all of which significantly increased the hydrogen yield of straw.

Reith et al. [9] provided an extensive list of heterotrophic bacteria that have been reported to produce hydrogen, including strict anaerobes such as *Clostridia*, facultative anaerobes such as *Enterobacter* and *Escherichia*, and aerobes such as *Alcaligenes* and *Bacillus*. In general, the aerobic metabolic rate is higher than the anaerobic metabolic rate, and growth rates of bacterial growth rates are higher under aerobic conditions than under strictly anaerobic conditions. This is

* *Corresponding author*. No. 9 Section 4, Renmin Nan Road, Chengdu, Sichuan, PR China. Tel.: +86 2882890229; fax: +86 2882890233.
E-mail address: lxf3636@163.com (X. Liu).
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because highly proliferative bacteria tend to produce more extracellular hydrolytic enzymes, such as cellulase and hemicellulase. According to the enzyme adsorption dynamics model proposed by South et al. [10], this increased enzyme concentration can improve the hydrolysis of lignocellulosic material. Therefore, appropriate oxygen addition can theoretically promote straw hydrolysis and further enhance hydrogen production when a mixed culture containing strict anaerobes, facultative anaerobes, and aerobes is used as the inoculum. Moreover, excess oxygen should inhibit strict anaerobes to produce hydrogen. However, this hypothesis has not yet been tested. In this study, the feasibility of micro-aerobic fermentative hydrogen production from corn straw (CS) was evaluated by a bottle test. Furthermore, the effect of oxygen addition on fermentative hydrogen production from straw was investigated using a bench-scale experiment.

Materials and methods

Substrates and inoculum

For the CS, air-dried CS was obtained from rural Chengdu, China. The collected CS was chopped and ground into small particles that were less than 1 mm in size for use as substrate. The total solid (TS) was 912.5 g/kg, and the volatile solid (VS), crude protein, crude fat, nitrogen free extract content was 91.8%, 3.8%, 0.4%, and 5.2% based on TS.

For the inoculum, the digested residue from a biogas reactor fed with pig manure was used as the original inoculum. The original inoculum was heated at 100 °C for 15 min and then cooled to room temperature in order to inhibit hydrogen-consuming bacteria [11]. The heated inoculum with pH 4.8, 1.52% TS, and 0.75% VS was used as the inoculum for fermentative hydrogen production in this study.

The nutrition solution contained 80 g/L NH_4HCO_3, 12.4 g/L KH_2PO_4, 0.1 g/L $MgSO_4{\cdot}7H_2O$, 0.01 g/L NaCl, 0.01 g/L $NaMoO_4{\cdot}2H_2O$, 0.01 g/L $CaCl_2{\cdot}2H_2O$, 0.015 g/L $MnSO_4{\cdot}7H_2O$, and 0.0278 g/L $FeCl_3$.

Experimental design

Bottle test

The CS (3.2 g based on TS), inoculum (24 mL), the nutrient solution (4 mL), and water (48.8 mL) were added to 150-mL anaerobic bottles with initial substrate concentrations of 40 g TS/L. Initial pH of the mixed materials was adjusted to 7.0 with 1 M NaOH solution for all bottles. The bottles were purged with N_2 to create an anaerobic condition and sealed with a butyl rubber stopper. The bottles were placed in a shaking incubator at 35 °C and 120 rpm for 72 h. The oxygen addition of 0, 0.70, 0.14, 0.21, 0.28, or 0.35 mL/(g TS·day) was carried out using a syringe. Each oxygen addition test was run in three parallel bottles.

Bench-scale test

The scale-up experiment of micro-aerobic fermentative hydrogen production was carried out in a bioreactor with a total volume of 10 L (Fig. 1S). Based on the results of the bottle test, the oxygen addition of 0.28 mL/(g TS·day) was selected for the bench-scale test. The CS (320 g based on TS), the mixed inoculum (2.4 L), the nutrient solution (0.4 L), and water (4.88 L) were added to the fermenter. The initial pH of the mixed materials was adjusted to 7.0 with the NaOH solution. The fermenter was operated at 35 °C for 72 h. As a comparison, anaerobic fermentation was carried out in another bioreactor without oxygen addition.

Analysis

TS and VS were determined according to standard methods [12]. The pH and oxidation reduction potential (ORP) were monitored using an online pH (HA405-DXK) and OPR meter (PT4805-60) (Mettler Toledo, Switzerland) which was equipped to bioreactor. Lignocellulose were determined using the Van Soest method [13].

Biogas production was estimated by measuring the water displacement in the case of the batch bottle test and by using the LML-1 wet gas meter (Changchun Automobile Filter Co., Ltd, China) in the bench experiments. Biogas analysis was performed using an Agilent 6890 gas chromatography (GC) system (Agilent Technologies, USA) with a thermal conductivity detector (TCD) and a 2-m stainless steel column packed with Porapak Q (50/80 mesh). The operating temperatures at the injection port, column oven, and detector were 100, 70, and 150 °C, respectively. Argon at a flow rate of 30 mL/min was used as the carrier gas.

Liquid samples were centrifuged at 12,000 rpm for 10 min at room temperature and filtered with a 0.45-μm membrane filter for analyzing the alkalinity, and volatile fatty acid (VFA) concentrations. Alkalinity was determined using the ZDJ-4B automatic potentiometric titrator (Shanghai Inesa Instrument Co., Ltd., China) by the titration method [14]. Concentrations of the VFA (including acetate, propionate, butyrate, iso-butyrate, valerate, and iso-valerate) in the supernatant were measured after acidification with 6 mol/L of HCl using an Agilent 6820 GC (Agilent Technologies, USA) equipped with a flame ionization detector (FID) and a 30 m × 0.25 mm × 0.25 μm capillary column (DB-FFAP). The temperatures at the injection port and detector were 250 and 300 °C, respectively. The initial temperature of the column oven was maintained at 50 °C for 5 min, after which the temperature was ramped at a rate of 10 °C/min until a final temperature of 250 °C was reached. The final temperature was maintained for 5 min. Nitrogen at a flow rate of 1 mL/min was used as the carrier gas [11]. Lactic acid was quantified using a high-performance liquid chromatograph (Waters 2695, USA) with a refractive index (RI) 2414 detector and a Shodex KC-811 S-DVB gel Column. 0.1%H_3PO_4 was used as the mobile phase at a flow rate of 0.7 mL/min at 50 °C [15].

Kinetic modeling

A modified Gompertz equation (Eq. (1)) was employed to model the process of hydrogen production [11]:

$$H(t) = P\exp\left\{-\exp\left[\frac{R_m e}{P}(\lambda - t) + 1\right]\right\} \quad (1)$$

Where H is the cumulative hydrogen production (mL/gVS) at fermentation time t (h), P is the hydrogen production

potential (mL/gVS), R_m is the maximum hydrogen production rate (mL/(gVS · h), e = 2.71828, and λ is the lag-phase time (h). The values of P, R_m, and λ were obtained from exponential regression analysis using Curve Expert, version 1.3.

Eq. (1) describes cumulative hydrogen production. The hydrogen production rate ν(mL/(gVS • h) can be described by differentiating the modified Gompertz equation (Eq. (2)).

$$\nu(t) = \frac{dH(t)}{dt} = R_m \exp\left\{2 + \frac{R_m e}{P}(\lambda - t) - \exp\left[\frac{R_m e}{P}(\lambda - t) + 1\right]\right\} \quad (2)$$

And t_{max} (the time of ν achieving ν_{max}) can be calculated as (Eq. (3)).

$$t_{max} = \lambda + \frac{P}{R_m e} \quad (3)$$

DNA isolation and PCR-DGGE analysis

Total genomic DNA was extracted from sludge collected at different fermentation time using a modified phenol/chloroform/isopropanol method [16]. For the PCR amplification, the universal primer set—including forward primer, 341F (5′- CCTACGGGAGGCAGCAG -3′), and reverse primer, 518R (5′- ATTACCGCGGCTGCTGG -3′), and GC clamp (5′- CGCCCGCCGCGCGCGGCGGGCGGGGCGGGGGCACGGGGGG -3′)—were used for the PCR amplification of the variable V3 region of 16S rDNA from the purified genomic DNA. PCR amplification was conducted in an automated thermal cycler using the following protocol, that is; initial denaturation for 4 min at 94 °C, 30 cycles of denaturation for 1 min at 94 °C, annealing for 30 s at 55 °C, extension for 1 min at 72 °C, followed by a final extension for 10 min at 72 °C.

Denaturing gradient gel electrophoresis (DGGE) of PCR products was performed with a DCode universal mutation system (Bio-Rad laboratories, Hercules, CA, USA). The DGGE analysis of PCR products was performed by electrophoresis for 10 min at 200 V and 12 h at 90 V through a 7.5% polyacrylamide gel containing a linear gradient of denaturant ranging from 45% to 65% in 1 × TAE buffer at a constant temperature of

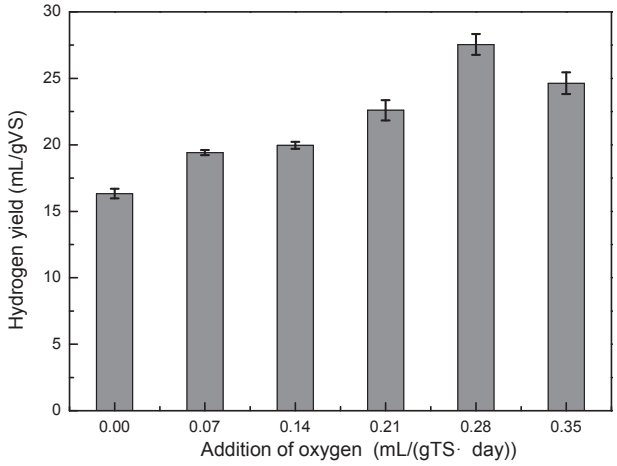


Fig. 1 – Hydrogen yields of corn straw with different amounts of oxygen addition.

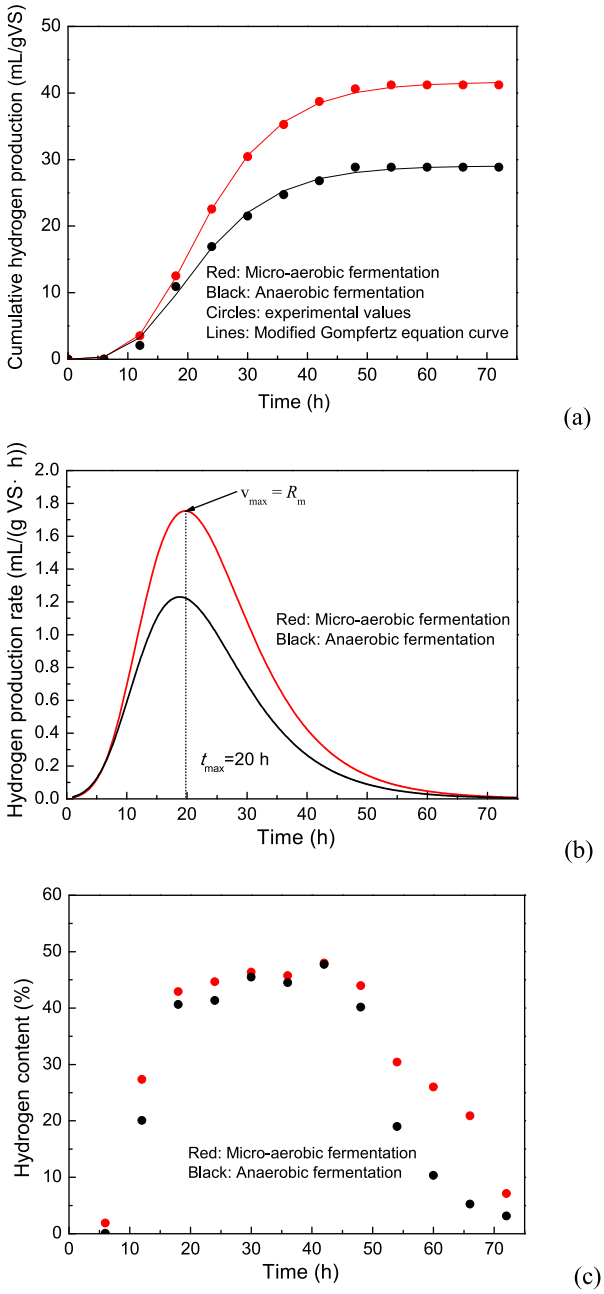


Fig. 2 – (a) Cumulative hydrogen production, (b) hydrogen production rate, and (c) hydrogen content during fermentative hydrogen production. The circle symbols represent the experimental value for micro-aerobic fermentation (red) and anaerobic fermentation (black), and the lines represent the modified Gompfertz equation curve.

Table 1 – Lignocellulose components of corn straw.

Component	Raw corn straw	After micro-aerobic fermentation	After anaerobic fermentation
Hemicellulose (%)	26.06 ± 0.68	12.78 ± 0.43	15.52 ± 0.38
Cellulose (%)	45.12 ± 1.19	33.06 ± 1.02	36.24 ± 0.97
Lignin (%)	10.86 ± 0.23	12.26 ± 0.36	11.4 ± 0.25

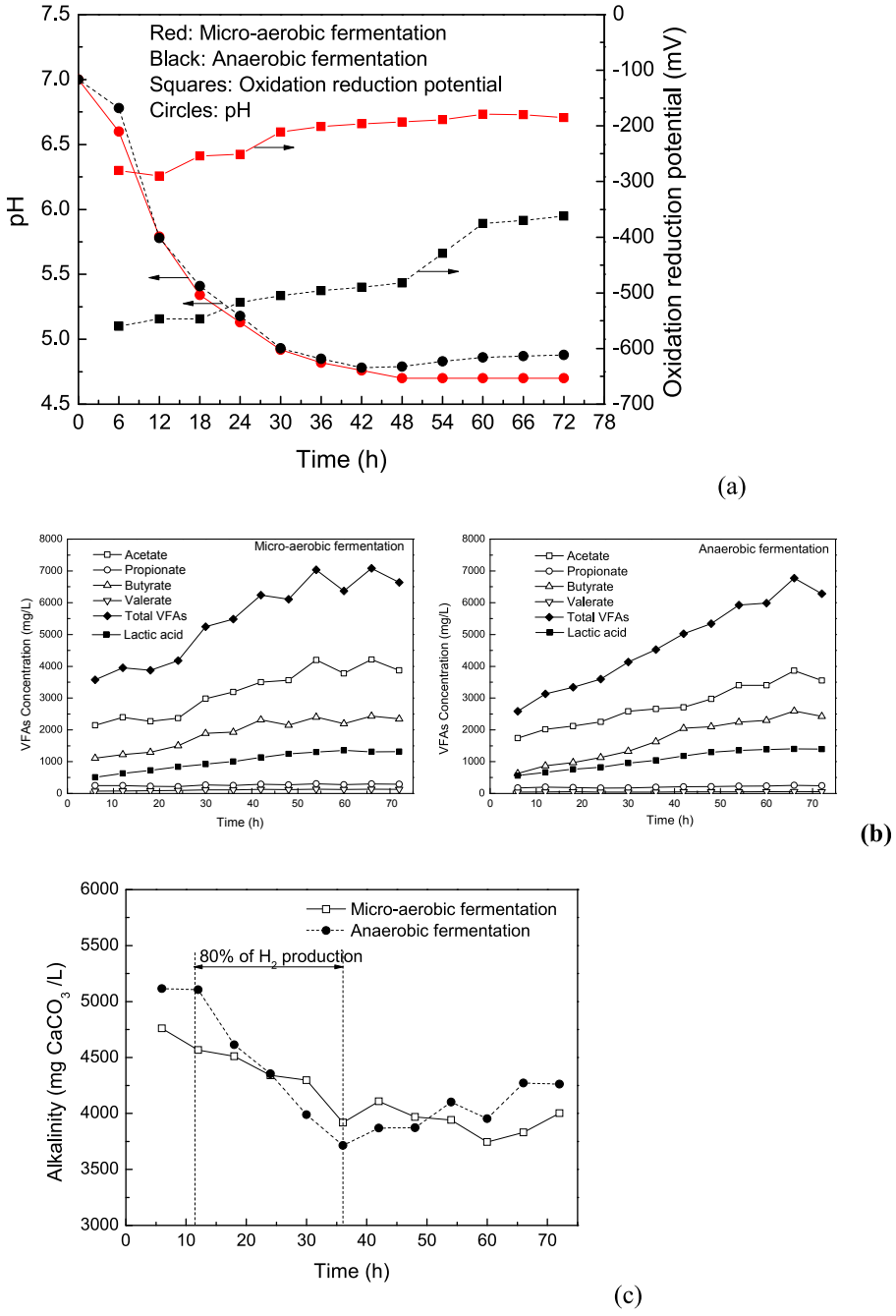


Fig. 3 – (a) Evaluation of pH and oxidation reduction potential (ORP), (b) analysis of volatile fatty acids (VFA) and (c) alkalinity during fermentative hydrogen production.

60 °C. The gel was stained with silver staining for 15 min and visualized on a UV transilluminator. The fingerprint of DGGE was analyzed using Quantity One V4.62 (Bio-Rad laboratories, Hercules, CA, USA).

Most of the bands were excised from the gel and re-amplified with the forward primer without a GC clamp. After re-amplification, the PCR products were cloned using TOP 10 competent cell (Tiangen Biotech, China) and pMD TM 19 – T Vector cloning kit (TaKaRa, China) in accordance with the manufacturer's instructions.

The sequences of re-amplified and cloned DNA fragments were determined by dideoxy chain termination (Genewiz, Suzhou, China) and compared with available sequences in GenBank database using the BLAST program. CLUSTAL X was used to align obtained sequences with sequences of reference microorganisms retrieved from GenBank. A phylogenetic tree was then constructed using the neighbor-joining method with MEGA 6.0. Bootstrapping analysis for 1000 replicates was performed to estimate the confidence of tree topologies.

Results and discussion

Determination of optimal oxygen addition

Hydrogen yields of bottle test with different amounts of added oxygen are shown in Fig. 1. A maximum hydrogen yield of 27.5 mL/g VS was obtained for oxygen addition of 0.28 mL/(g TS·day), which was 68.71% higher than that without oxygen addition. These results indicated that the optimal oxygen addition is 0.28 mL/(g TS·day) in micro-aerobic fermentative hydrogen production.

Comparison on hydrogen production and lignocellulose degradation

The modified Gompertz equation was used to appropriately model the process of hydrogen production with a correlation coefficient (R^2) of 0.9998 and 0.9986 for scale-up micro-aerobic and anaerobic fermentative hydrogen production, respectively. In addition to improving the hydrogen yield, the micro-aerobic condition increased the hydrogen production rate and increased hydrogen content in the later period (Fig. 2). The positive effect of moderate oxygen addition is likely due to improvement in the activity of facultative anaerobes or aerobes, which in turn increased the production of extracellular hydrolytic enzymes. However, the cellular and enzymatic mechanism of micro-aerobic fermentative hydrogen production requires further study. It is worth mentioning that the hydrogen yield of 41.6 mL/g VS from bench-scale test was much greater than that of 27.5 mL/g VS from bottle test. This may be due to the good mixing condition of scale-up bioreactor. Therefore, there is space for micro-aerobic fermentative hydrogen production through straw pretreatment, bioreactor and process optimization.

The hemicellulose, cellulose, lignin, and ash contents of raw CS and fermented residues are listed in Table 1. Hydrogen production fermentation significantly reduced the contents of hemicellulose and cellulose, which are easily degradable. The reduction of hemicellulose and cellulose was greater in micro-aerobic fermentation than anaerobic fermentation, which might be due to the enhanced hydrolysis resulting from the oxygen addition.

Comparison on process parameters and by-products

The process parameters of ORP, pH, and alkalinity, and the by-products of VFA were monitored during the process of fermentative hydrogen production (Fig. 3). The ORP of micro-aerobic fermentation (−300 to −150 mV) was clearly higher than that of anaerobic fermentation (−560 to −360 mV), due to the addition of oxygen. For micro-aerobic fermentation, it is understandable that redox potential increased to higher value through the fermentation, since oxygen was added everyday. For anaerobic fermentation, some oxidative substances should be produced in system in order to meet the redox balance, since the reducing hydrogen was output from system. However, potential oxidative substances need to be confirmed in further study. The oxidized form of nicotinamide-adenine dinucleotid (NAD^+) could be the one of main oxidative substances (Eq. (4)) [17].

$$NADH + H^+ \rightarrow NAD^+ + H_2 \quad \Delta G^{0\prime} = -206.3 kJ/mol \tag{4}$$

VFA are by-products of fermentative hydrogen production (Eqs. (5) and (6)). The VFA concentration of micro-aerobic fermentation (maximum 7085 mg/L) was higher than that of anaerobic fermentation (maximum 6773 mg/L) for the entire hydrogen production process. For both micro-aerobic and anaerobic fermentative hydrogen production, the main VFA were acetate and butyrate, which accounted for more than

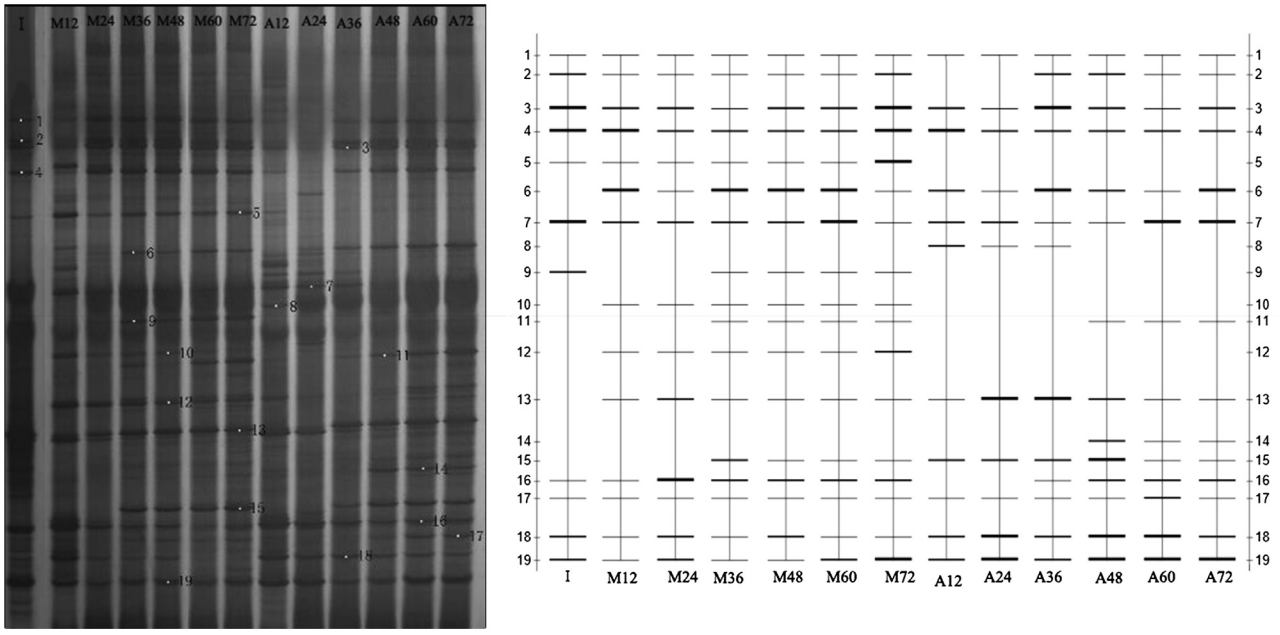


Fig. 4 – Fingerprint (left) and band intensity (right) of denaturing gradient gel electrophoresis. I: inoculum, M: micro-aerobic fermentation, A: anaerobic fermentation, Number: fermentation time (h).

90% of the total VFA. It can be concluded that both the micro-aerobic and anaerobic hydrogen production were butyric acid-type fermentation [18].

$$C_6H_{12}O_6+2H_2O\rightarrow 2CH_3COOH+2CO_2+4H_2 \quad \Delta G^{0\prime}=-206.3kJ/mol \tag{5}$$

$$C_6H_{12}O_6\rightarrow CH_3CH_2CH_2COOH+2CO_2+2H_2 \quad \Delta G^{0\prime}=-254.8kJ/mol \tag{6}$$

VFA accumulation decreases pH value. For anaerobic fermentation, the pH value decreased from 7.0 to 4.9 with accumulation of VFA, whereas it decreased to 4.7 in micro-aerobic fermentation (Fig. 3a). While more VFA was produced with micro-aerobic fermentation, the pH of both fermentations was similar during the period from 12 h to 36 h. In general, the pH of the fermentation system was controlled by a buffer system, and alkalinity was used to quantify the buffering capacity. Fig. 3c shows the alkalinity obtained. An obvious difference between the micro-aerobic and anaerobic fermentation was observed for alkalinity during the period of 12–36 h, where 80% of the total hydrogen was produced. The absolute reduction of alkalinity was 648 mg $CaCO_3$/L for micro-aerobic fermentation, and it was 1390 mg $CaCO_3$/L for anaerobic fermentation.

A modified buffer index (*BI*) is presented as Eq. (7) to evaluate the buffer ability of fermentative hydrogen production system based on the traditional buffer index [19]. The lower *BI* value indicates a strong buffer ability of fermentation system. Here, the modified buffer index of micro-aerobic fermentation and anaerobic fermentation were 668 and 1497. It seems that the micro-aerobic condition had strong buffer ability to tolerate higher VFA concentrations, which was helpful to produce more hydrogen.

$$BI=\frac{\Delta A}{\Delta pH} \tag{7}$$

Where ΔA is the difference value between final and initial alkalinity of fermentative system, ΔpH is the difference value between final and initial pH of fermentative system, *BI* is the modified buffer index that is defined as a positive dimensionless value.

Comparison on bacterial community

An image of a DGGE gel of the PCR-amplified 16S rDNA segments extracted from the inoculum (Lane I), sludge sampled from micro-aerobic fermentation (Lane M12 – M72) and anaerobic fermentation (A12 – A72) at different fermentation time is depicted in Fig. 4 (left). And corresponding band intensity of DGGE is shown in Fig. 4 (right). The differences of band numbers and band intensity were observed between micro-aerobic and anaerobic fermentation. The number of bands detected from the micro-aerobic condition was greater than that from the anaerobic condition, suggesting microbial community of micro-aerobic fermentation was more abundant than that of anaerobic fermentation.

The Shannon-wiener index was used to evaluate the diversity of microbial community. The greater Shannon-wiener index presents more abundant microbial community. The Pielou index was used to evaluate the distribution evenness of microbial community. The higher Pielou index indicates the more even distribution of microbial species. The Shannon-wiener and Pielou index were shown in Fig. 5. At the same fermentation time, the diversity of microbial community under micro-aerobic condition was richer than that under anaerobic condition. On the contrary, distribution evenness of microbial community under micro-aerobic condition was poorer than that under anaerobic condition. This clearly indicates that the dominant position of some bacteria in micro-aerobic system is more obvious, even though the microbial species of micro-aerobic system was more than that of anaerobic system.

To identify microorganisms corresponding to bands 1–19, the bands were excised and purified to determine their 16S rDNA sequences. A phylogenetic tree was then constructed by comparing the obtained sequences with database sequences using the BLAST search. The functions related to hydrogen production of these bacteria were shown in Table 2. Bacteria of *Enterococcus* sp. (Band 3), *Enterococcus casseliflavus* (Band 7),

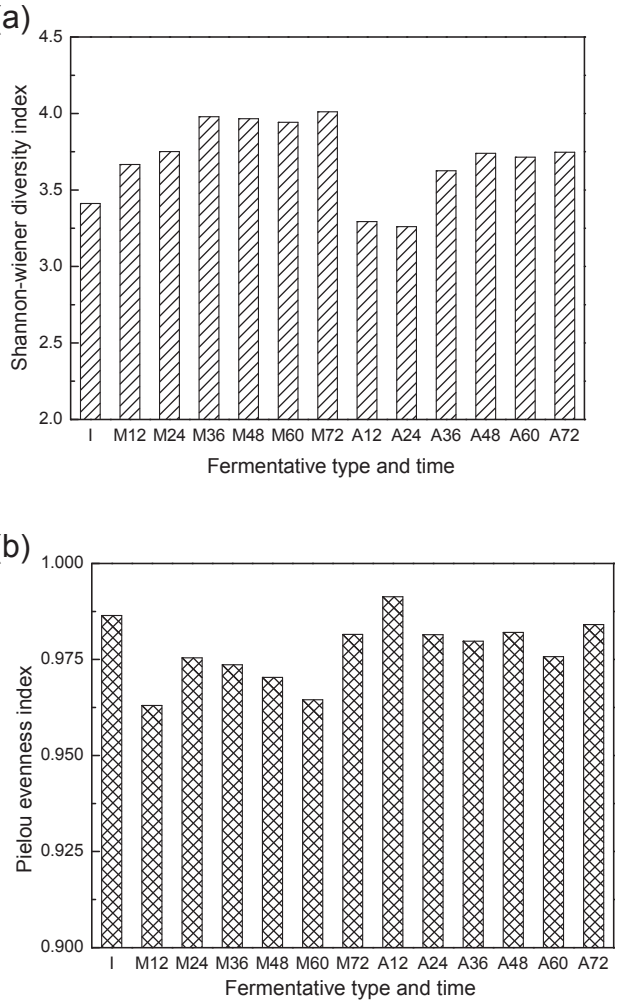


Fig. 5 – Analysis of (a) diversity index and (b) evenness index of fermentative bacterial system. I: inoculum, M: micro-aerobic fermentation, A: anaerobic fermentation, Number: fermentation time (h).

Table 2 – Bacteria in inoculum, micro-aerobic and anaerobic fermentation system.

Band	Bacteria	Similarity	Inoculum	Micro-aerobic	Anaerobic	Function	Ref.
Band 1	Acinetobacter sp. SG-3	100%	+	+	+[a]	Producing cellulolytic hydrolase (mainly xylanase)	[20]
Band 4	*Bacteroides oleiciplenus* YIT 12058	100%	+	+	+	Producing xylanase or Fermenting oil to hydrogen	[25,26]
Band 8	*Bifidobacterium animalis*	100%	–	–	+[b]	Fermenting monose to lactate and acetate without hydrogen production	
Band 16	*Bifidobacterium animalis*	100%	+	+	+[a]		
Band 19	*Bifidobacterium animalis*	100%	+	+	+		
Band 2	*Clostridium butyricum*	100%	+	+	+[a]	Fermenting mono- and disaccharides to acid and hydrogen	[20]
Band 18	*Clostridium baratii*	100%	+	+	+		[27,28]
Band 14	*Clostridium septicum*	99%	–	–	+[b]	Unknown for hydrogen production, Pathogens	[29]
Band 5	Citrobacter sp. enrichment culture	99%	+	+	–	Fermenting glucose to acid and hydrogen	[22]
Band 7	*Enterococcus casseliflavus*	97%	+	+	+	Degrading cellulose and producing acid and hydrogen	[30]
Band 3	*Enterococcus* sp. ZGZ2	100%	+	+	+		
Band 12	*Escherichia* sp. ESBL525B1	100%	–	+	–	Producing hydrogen form formate by formate hydrogen lyase	[31]
Band 13	*Lactobacillus mucosae*	100%	–	+	+	Fermenting monose to lactate without hydrogen production	
Band 15	*Lactobacillus mucosae*	99%	–	+[c]	+		
Band 6	*Peptoclostridium difficile* (*Clostridium difficile*)[d]	99%	–	+	+	Fermenting mono- and disaccharides to acid and hydrogen	[32]
Band 9	Uncultured Bacteroidetes bacterium	100%	+	+[c]	–	/	
Band 17	Uncultured Clostridia bacterium	99%	+	+	+	/	

[a] No observation during the early stage (0–24 h) of anaerobic fermentation.
[b] No observation during the early stage (0–36 h) of anaerobic fermentation.
[c] No observation during the early stage (0–24 h) of micro-aerobic fermentation.
[d] Synonym, name used before.

Bacteroides oleiciplenus (Band 4), *Peptoclostridium difficile* (Band 6), *Lactobacillus mucosae* (Band 13 and 15), *Clostridium butyricum* (Band 2),*Clostridium baratii* (Band 18), *Bifidobacterium animalis* (Band 16 and 19), and an uncultured *Clostridia* bacterium (Band 17) existed in micro-aerobic and anaerobic fermentation system. The *Clostridia* group is well-known to produce acetic acid, n-butyric acid and hydrogen via dark fermentation based on Eqs. (4) and (5) [20]. *Citrobacter* sp. (Band 5), *Escherichia* sp. (Band12), and an uncultured Bacteroidetes bacterium (Band 9) existed in the micro-aerobic system, but not in the anaerobic system. *Citrobacter* sp. and *Escherichia* sp. were reported facultative anaerobes producing hydrogen [21,22]. Promotion of micro-aerobic condition is likely due to these bacteria. Some bacteria were not detected in inoculum could come from straw material.

Existence of *B. animalis* (Band 8, 16 and 19) and *L. mucosae* (Band 13 and 15) might be responsible for the low hydrogen production. These microorganisms can utilize carbohydrate-rich substrate for the production of lactic acid, instead of hydrogen [23], and other metabolites such as bacteriocins that can inhibit the activity of hydrogen producing bacteria [24]. There are two categories of lactic acid bacteria (LAB) i.e. homofermentative bacteria (*Lactococcus* and *Lactobacillus*) and heterofermentative bacteria (*Leuconostoc* and *Bifidobacterium*). Lactic acid is the only end product of glucose metabolism by homofermentative bacteria through the Embden-Meyerhof-Parnas (EMP) pathway as showed in the Eq. (8). Lactic, acetic acid and CO_2 are the end products of glucose metabolism by heterofermentative bacteria through the bifidus pathway as showed in the Eq. (9). The detected maximum lactic acid concentrations were 1352 mg/L and 1400 mg/L for micro-aerobic fermentation and anaerobic fermentation (Fig. 3b).

$$C_6H_{12}O_6 + 2ADP + 2Pi \rightarrow 2CH_3CH(OH)COOH + 2ATP \quad (8)$$

$$C_6H_{12}O_6 + 2ADP + 2Pi + 2NAD^+ \rightarrow CH_3CH(OH)COOH + CH_3COOH + 2CO_2 + 2NADH + H^+ + 2ATP \quad (9)$$

Therefore, *B. animalis* and *L. mucosae* that ferment monose to lactate and acetate independent of hydrogen production should be removed or inhibited in order to increase hydrogen production. In especial, the removal or inactivation of indigenous microorganism *L. mucosae* existed in straw need to be further studied.

Conclusion

The micro-aerobic condition was an effective method for enhancing fermentative hydrogen production of straw. The hydrogen yield of 41.6 mL/g VS obtained from bench-scale test with oxygen addition of 0.28 mL/(g TS·day) was higher than that of 28.9 mL/g VS without oxygen addition. The oxidation reduction potential was −300 ~ −150 mV for micro-aerobic fermentation. Both the micro-aerobic and anaerobic hydrogen productions were butyric acid-type fermentation. Besides of *Enterococcus*, *Bacteroides*, *Peptoclostridium*, *Clostridium* genera that existed both in micro-aerobic and anaerobic fermentation system, *Citrobacter* sp. and *Escherichia* sp. were unique to the micro-aerobic system. The presence of LAB might lower fermentative hydrogen production.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.ijhydene.2016.01.141.

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